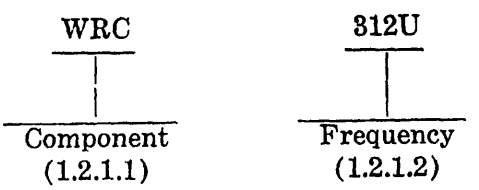
MILITARY SPECIFICATION

WAVEGUIDES, RIGID, CIRCULAR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.** This specification covers circular, rigid waveguides as fabricated in bulk at the mill. Waveguides may be seamless or laminated stock supplied in aluminum alloy, brass, copper, magnesium-base alloy, or silver alloy. The term "circular" applies to both the inside and outside geometry (see 6.1).



1.2.1.2 *Frequency.* The frequency is identified by a three-digit number and a letter. The number identifies the minimum operating frequency in the TE_{11} mode in megacycles, and the letter identifies a multiplier in accordance with table I.

TABLE I. *Letter multiplier.*

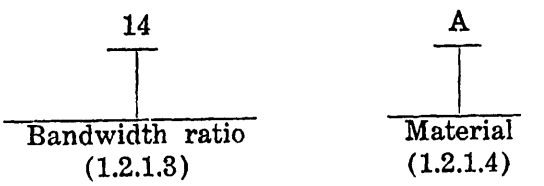
Symbol	Multiply by
U	1
D	10
C	100
K	1,000

1.2.1.3 *Bandwidth ratio.* The nominal bandwidth ratio is identified by the two-digit number "14". The number is the nominal bandwidth ratio multiplied by 10.

1.2 Classification.

1.2.1 *Type designation.* The type designation shall be in the following form, and as specified (see 6.2):

1.2.1.1 *Component.* Circular, rigid waveguides are identified by the three-letter symbol WRC.



1.2.1.4 *Material.* The material is identified by a one-letter symbol in accordance with table II.

TABLE II. *Material.*

Symbol	Material
A	Aluminum alloy
B	Brass
C	Copper
M	Magnesium-base alloy
S	Silver alloy

FSC 5985

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–A–561 — Aluminum Alloy 1100 (2S); Plate and Sheet.

PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL–P–116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue, at 33rd Street, New York 16, N. Y.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

B75–59 — Seamless Copper Tube.

B135–60 — Seamless Brass Tube.

B217–60 — Magnesium-Base Alloy Extruded Tubes.

B235–59T — Aluminum-Alloy Extruded Tubes.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material. The material shall be aluminum alloy, brass, copper, magnesium-alloy, or silver alloy, as specified (see 1.2.1.4 and 6.2). Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.1.1 *Aluminum alloy.* Aluminum alloy shall conform to temper F of Specification QQ–A–561, or alloy 6063 or 6061 of ASTM Publication B235-59T.

3.1.2 *Brass.* Brass shall conform to alloy 7, drawn (general purpose), of ASTM Publication B135–60.

3.1.3 *Copper.* Copper shall conform to type DLP of ASTM Publication B75–59.

3.1.4 *Magnesium-base alloy.* Magnesium-base alloy shall conform to composition AZ31B of ASTM Publication B217–60.

3.1.5 *Silver alloy.* Silver alloy shall be made of commercial coin silver having a

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composition of 90±0.1 percent silver and 10±0.5 percent copper.

3.2 **Design and construction.** Waveguides shall be of the design, construction, and physical dimensions specified (see table III).

3.2.1 *Finish.* The interior and exterior surfaces of the finished waveguide shall be bright, smooth, dry, and free of scales of oxide.

3.2.2 *Stock lengths.* Waveguides shall normally be supplied in the nominal stock lengths shown in table IV, as specified (see 6.2). When the nominal length is not specified, waveguides shall be furnished in 14-foot nominal stock lengths. Lengths shorter than the nominal length may be supplied, but in no case shall the total of such lengths be more than 25 percent of the total length of waveguides delivered on a contract or order, nor shall any single length of waveguide be shorter than the applicable acceptable length shown in table IV. Precise lengths shall be supplied when specified (see 6.2).

TABLE IV. *Stock lengths of waveguides.*

Nominal length	Tolerance	Acceptable length
Feet	*Inches* (+)	*Feet*
14	2	8
12	2	8
10	2	6
8	2	6
6	2	4
4	2	3
2	1	1

3.3 **Bow.** When measured as specified in 4.6.2, the bow, for any orientation, shall be not more than 0.010 inch between any two points 2 feet apart on the concave external surface of the finished waveguide (see 6.3.1).

3.4 **Eccentricity.** Eccentricity shall not exceed 10 percent (see 6.3.4).

3.5 **Marking.** Each length of finished waveguide shall be marked in accordance with Standard MIL-STD-130, with the type designation and the manufacturer's code symbol. The manufacturer's name or trademark may also be marked on the waveguide, provided such letters are not expressly forbidden in the contract or order, and their height does not exceed 50 percent of the outside diameter of the waveguide. The markings shall be applied with a permanent ink or paint. Not more than 6-inch intervals shall be permitted from the end of one marking to the beginning of the next marking.

3.6 **Workmanship.** The material shall be free from defects of a nature that may interfere with normal commercial applications. The finished waveguide shall be uniform in composition and wall thickness, straight and smooth from end to end, and shall be free from internal and external mechanical imperfections, in accordance with good commercial practice. In addition, the interior surface of the waveguide shall be free from burrs, plug marks, chatter marks, dirt, grease, scales, and splinters. Scratches not more than 1 mil deep will be permitted in the longitudinal direction for waveguides of 0.594-inch inside diameter or larger. For waveguides of less than 0.594-inch inside diameter, the scratches shall not exceed 0.12 percent of the diameter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.**

4.1.1 *Supplier.* The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE III. *Frequency ranges and dimensions for circular, rigid waveguides.*

Type designation [1]	Frequency range (kmc/s) [2]		Dimensions (inches)					
	TE01 mode	TE11 mode	Inside		Outside [3]		Nominal wall thickness [3]	Max. inside diameter out of roundness (see 6.3.3)
			Diameter (see 6.3.2)	Tolerance (±)	Diameter	Tolerance (±)		
WRC312U14-	0.683 — 0.940	0.312 — 0.427	25.508	0.025	...	...	...	0.025
WRC365U14-	0.799 — 1.100	0.365 — 0.500	21.791	0.020	...	...	...	0.022
WRC427U14-	0.936 — 1.290	0.427 — 0.586	18.616	0.020	...	...	...	0.019
WRC500U14-	1.100 — 1.510	0.500 — 0.686	15.908	0.015	...	...	...	0.016
WRC586U14-	1.280 — 1.770	0.586 — 0.803	13.585	0.015	...	...	...	0.014
WRC686U14-	1.500 — 2.070	0.686 — 0.939	11.606	0.010	...	...	...	0.012
WRC803U14-	1.760 — 2.420	0.803 — 1.100	9.915	0.010	...	...	...	0.010
WRC939U14-	2.060 — 2.830	0.939 — 1.290	8.470	0.008	...	...	...	0.008
WRC110D14-	2.410 — 3.310	1.100 — 1.510	7.235	0.007	...	...	...	0.007
WRC129D14-	2.820 — 3.880	1.290 — 1.760	6.181	0.006	6.781	0.008	0.300	0.006
WRC151D14-	3.300 — 4.540	1.510 — 2.070	5.280	0.005	5.880	0.007	0.300	0.006
WRC176D14-	3.860 — 5.320	1.760 — 2.420	4.511	0.005	5.111	0.007	0.300	0.005
WRC207D14-	4.520 — 6.220	2.070 — 2.830	3.853	0.004	4.253	0.006	0.200	0.005
WRC242D14-	5.290 — 7.280	2.420 — 3.310	3.292	0.003	3.692	0.005	0.200	0.005
WRC283D14-	6.190 — 8.530	2.830 — 3.880	2.812	0.003	3.112	0.005	0.150	0.005
WRC331D14-	7.250 — 9.980	3.310 — 4.540	2.403	0.0025	2.703	0.004	0.150	0.005
WRC389D14-	8.510 — 11.700	3.890 — 5.330	2.047	0.002	2.307	0.004	0.130	0.005
WRC454D14-	9.950 — 13.700	4.540 — 6.230	1.750	0.0015	2.010	0.004	0.130	0.004
WRC530D14-	11.600 — 16.000	5.300 — 7.270	1.500	0.0015	1.700	0.003	0.100	0.004
WRC621D14-	13.600 — 18.700	6.210 — 8.510	1.281	0.0015	1.441	0.003	0.080	0.004
WRC727D14-	15.900 — 21.900	7.270 — 9.970	1.094	0.0013	1.224	0.003	0.065	0.004
WRC849D14-	18.600 — 25.600	8.490 — 11.600	0.938	0.0013	1.068	0.003	0.065	0.004
WRC997D14-	21.900 — 30.100	9.970 — 13.700	0.797	0.0013	0.897	0.0025	0.050	0.003
WRC116C14-	25.300 — 34.900	11.600 — 15.900	0.688	0.0013	0.788	0.0025	0.050	0.003
WRC134C14-	29.300 — 40.400	13.400 — 18.400	0.594	0.001	0.674	0.0025	0.040	0.003
WRC159C14-	34.800 — 48.000	15.900 — 21.800	0.500	0.001	0.580	0.002	0.040	0.003
WRC182C14-	39.800 — 54.800	18.200 — 24.900	0.438	0.001	0.518	0.002	0.040	0.003
WRC212C14-	46.400 — 63.900	21.200 — 29.100	0.375	0.001	0.435	0.002	0.030	0.003
WRC243C14-	53.100 — 73.100	24.300 — 33.200	0.328	0.001	0.388	0.002	0.030	0.002
WRC283C14-	61.900 — 85.200	28.300 — 38.800	0.281	0.001	0.341	0.002	0.030	0.002
WRC318C14-	69.700 — 95.900	31.800 — 43.600	0.250	0.001	0.290	0.002	0.020	0.002
WRC364C14-	79.600 — 110.000	36.400 — 49.800	0.219	0.001	0.259	0.002	0.020	0.002
WRC424C14-	92.900 — 128.000	42.400 — 58.100	0.188	0.001	0.228	0.002	0.020	0.002
WRC463C14-	101.000 — 139.000	46.300 — 63.500	0.172	0.001	0.212	0.002	0.020	0.002
WRC566C14-	124.000 — 171.000	56.000 — 77.500	0.141	0.001	0.181	0.002	0.020	0.002
WRC635C14-	139.000 — 192.000	63.500 — 87.200	0.125	0.001	0.155	0.002	0.015	0.002
WRC727C14-	159.000 — 219.000	72.700 — 99.700	0.109	0.001	0.139	0.002	0.015	0.002
WRC848C14-	186.000 — 256.000	84.800 — 116.000	0.094	0.001	0.124	0.002	0.015	0.002

[1] Complete type designation shall include an additional letter symbol to indicate material.
[2] Frequency ranges are given for information only.
[3] Dimensions on outside diameter with tolerances and wall thickness are omitted on types WRC312U14- to WRC110D14-, inclusive, because manufacturing methods on these sizes may vary widely, depending on the individual application.

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4.1.1.1 *Test equipment and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 **Classification of inspection.** The examination and testing of waveguides shall be classified as follows:

(a) Component-materials inspection (see 4.3).

(b) Acceptance inspection (see 4.5).

1. Inspection of product for delivery (see 4.5.1).
2. Inspection of preparation for delivery (see 4.5.2).

4.3 Component-materials inspection. Material shall be inspected to determine conformance to contractually applicable specifications. Such inspection may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The inspection results shall be available to the Government inspector for review. Certified inspection reports, identifiable with the material, may be accepted in lieu of repeated inspections. The Government reserves the right, in any case, to require inspection to assure that the materials are acceptable. The sample selected shall be representative of the finished product.

4.4 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at room ambient temperature, relative humidity, and pressure.

4.5 Acceptance inspection.

4.5.1 *Inspection of product for delivery.* Inspection of product for delivery shall consist of groups A and B.

4.5.1.1 *Inspection lot.* An inspection lot, as far as practicable, shall consist of all the waveguides of the same type designation, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 *Nonconforming lots.* Nonconforming lots resubmitted for acceptance shall be inspected using tightened inspection (see applicable paragraph of Standard MIL-STD-105.)

4.5.1.3 *Group A inspection.* Group A inspection shall consist of visual and mechanical examination (see 4.6.1 and 4.6.1.1).

4.5.1.3.1 *Sampling plan.* Statistical sampling plan and inspection shall be in accordance with Standard MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be 1.0 and 4.0 (percent defective) for major and minor defects, respectively. Major and minor defects shall be as defined in Standard MIL-STD-105.

4.5.1.4 *Group B inspection.* Group B inspection shall consist of the test for bow (see 4.6.2).

4.5.1.4.1 *Sampling plan.* The sampling plan shall be in accordance with Standard MIL-STD-105 for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. For small-sample reduced inspection, procedure R-1 shall be used. The AQL shall be 6.5 (percent defective), and the inspection level shall be L7 for normal and tightened inspection, and L5 for reduced inspection.

4.5.1.4.2 *Disposition of sample units.* Sample units which have passed the group B inspection shall be delivered on the contract or order, if the lot is accepted.

4.5.2 *Inspection of preparation for delivery.* Sample items and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with requirements of section 5 of this specification.

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4.6 Methods of examination and test.

4.6.1 *Visual and mechanical inspection.* Waveguides shall be examined to verify that the design, construction, physical dimensions, eccentricity, marking, and workmanship are in accordance with the applicable requirements (see 3.2 to 3.2.2, incl., and 3.4 to 3.6, incl.).

4.6.1.1 *Dimensions.* Each waveguide shall be measured at a point at least 1 inch from the end for tubes of 1-inch outside diameter and larger, and at least ½ inch from the end for tubes of less than 1-inch outside diameter, to ascertain that all dimensions are as specified (see table III) and that the waveguide is of the specified length (see 3.2.2 and 6.2). Wall thickness may be measured at any point around the periphery; measurements may depart from the nominal, due to the wall being thicker or thinner than nominal, or to the waveguide being eccentric, or a combination of these conditions.

4.6.2 *Bow.* The waveguide shall be so positioned during measurement that gravity will not tend to increase the amount of bow. The bow shall be determined by using a straightedge (see 3.3).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

5.1.1 *Level A.* The ends of each length of waveguide shall be sealed to prevent the entrance of water vapor. The protective barrier shall extend at least 6 inches back from both ends. No further packaging will be required.

5.1.2 *Level C.* Lengths of waveguides shall be afforded preservation and packaging in accordance with the supplier's normal commercial practice.

5.2 Packing (see 6.2).

5.2.1 *Level A.* The lengths of waveguide shall be separated according to size and packed in nailed wood boxes conforming to Specification PPP–B–621. The lengths shall be kept separate by means of wood collar blocks, spaced not more than 24 inches apart, and secured to the box to prevent movement while in transit. The gross weight shall not exceed 150 pounds except when the weight of an individual length of waveguide exceeds 150 pounds.

5.2.2 *Level C.* Lengths of waveguides packaged as specified (see 6.2) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129 (see 6.2).

6. NOTES

6.1 Intended use. Waveguides covered by this specification are intended for use as transmission lines in Armed Services radio and electronic equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The complete type designation (see 1.2.1 and table III.)

(c) Stock lengths or precise lengths of waveguides required (see 3.2.2).

(d) Levels of preservation and packaging and packing, and applicable marking (see sec. 5).

6.3 Definitions. For the purpose of this

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specification, the following definitions shall apply:

6.3.1 *Bow.* Bow is the departure from a straight longitudinal line connecting any two points 2 feet apart on the external surface of the waveguide without the effect of gravity or any other forces.

6.3.2 *Nominal inside diameter.* Nominal inside diameter is the average at one cross section, of the maximum and minimum measured diameters usually found at, or very close to, 90° from each other. Nominal inside diameter =

$$\frac{\text{Maximum inside diameter} + \text{minimum inside diameter}}{2}$$

6.3.3 *Out of roundness.* Out of roundness is the total difference found at any cross section between the individual maximum and minimum inside diameter which usually occurs at, or about, 90° from each other. Out of roundness is not expressed in plus or minus. Out of roundness = Maximum inside diameter —minimum inside diameter.

6.3.4 *Eccentricity.* Eccentricity is the maximum deviation of the wall thickness at any cross section from the mean wall thickness at that cross section.

6.4 **Engineering information.**

6.4.1 *Frequency range.* The recommended frequency range of each waveguide size is listed in table III, and assumes operation in the dominant (TE_{11}) mode or in the circular electric (TE_{01}) mode. Each recommended frequency range for the TE_{01} mode operation is within 1.21 times the cutoff frequency for the TE_{01} mode and 0.91 times the cutoff frequency for the TE_{02} mode. Each recommended frequency range for the TE_{11} mode operation is within 1.15 times the cutoff frequency for the TE_{11} mode and 0.95 times the cutoff frequency for the TE_{21} mode. To permit selection of optimum waveguide sizes, two overlapping series are listed.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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